# Stick-together calculations

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Find a common final velocity using total momentum.

**SUCCESS CRITERIA - I can...**

1. Add the initial momenta with signs.
2. Use the combined mass after sticking.
3. Check the result against the direction of total momentum.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “inelastic” and “initial” to explain your answer. Sentence starters are optional.

**inelastic**: a collision that transforms some kinetic energy to other forms

**initial**: at the beginning of the interval; subscript i

**final**: at the end of the interval; subscript f

## 01 Warm-up: bring back an earlier idea

Two carts separate from rest with no external horizontal impulse. One has momentum **15 kg·m/s** left. **Find the other’s momentum**.

**Word help:** external = acting from outside the chosen system

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $v_f = (m_1 v_{1,i} + m_2 v_{2,i})/(m_1 + m_2)$.
Sticking collision in one dimension with negligible external impulse. $v_f$ is the shared final velocity.

**Worked reasoning:** A **2 kg** cart at **+3 m/s** sticks to a **1 kg** cart at rest. Total momentum **6 kg·m/s** divided by total mass **3 kg** gives final velocity **+2 m/s**.

**Your turn:** Mark the conserved quantity on a before-and-after sketch. **Explain why masses** are added in the final denominator.

**Sentence starter:** I think … . I can point to …, which shows …, so … .

**Before** **After**

Label mass, velocity and momentum.

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

Two carts stick. **Identify which masses belong** in the final momentum expression and why.

**Sentence starter:** I think … . I can point to …, which shows …, so … .

Before contact: add velocity arrows.




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A **1 kg** cart moves right at **6 m/s** and sticks to a **2 kg** cart at rest. **Find initial total momentum**, combined mass and final velocity. Organize the known quantities on the before-and-after diagram.

**Word help:** at rest = not moving in the chosen reference frame

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Before** **After**

Label objects and signed velocities.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Explain your reasoning

A student averages the two starting velocities without considering mass. **When would** that happen to work?

**Sentence starter:** I notice … . This supports … because … .

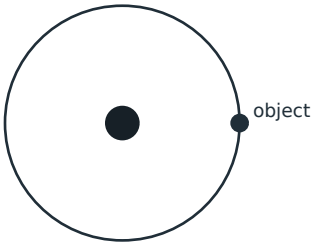
**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A car follows a circle at steady **6 m/s**. **Sketch velocity** at two points. Could net force be zero while it follows this path?
**Label velocity** and inward acceleration on the circular-path diagram.

**Word help:** net force = the vector sum of all forces on one object

**Sentence starter:** In my work, I used … to show … . I labeled it … .




**Marzano check - circle:** 1 2 3

## 07 Exit check

Two equal **1 kg** carts approach at **2 m/s** in opposite directions and stick. **Find final velocity**. Organize the known quantities on the before-and-after diagram.

**Sentence starter:** I found … [unit]. This value means … in this situation.

| Before | After |
| --- | --- |
| | |

Label objects and signed velocities.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

**Choose the mass of a** stationary cart so a **2 kg** cart at **6 m/s** slows to **2 m/s** when they stick. Organize the known quantities on the before-and-after diagram.

**Sentence starter:** I notice … . This supports … because … .

| Before | After |
| --- | --- |
| | |

Label objects and signed velocities.

**Marzano check - circle:** 1 2 3